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Analog counterdiabatic quantum computing

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Harnessing the analog capacity of quantum processors at the algorithmic level is key to solving computationally hard problems. Neutral atoms offer analog capabilities supporting hundreds of qubits, but state-of-the-art adiabatic protocol struggles with nonadiabatic errors, restricting scalability due to finite coherence times. To address this, we propose and experimentally demonstrate a tailored analog counterdiabatic quantum computing (ACQC) protocol to enhance the computational capabilities of neutral atoms by mitigating non-adiabatic transitions and facilitating rapid and high-quality solutions. We apply it to solve the maximum independent set problem with up to 100 qubits, achieving over 3-fold speedup in convergence time and solution quality within a short evolution time of the processor, as compared to adiabatic method. Our method shows scalability of the application of neutral atom processors establishing ACQC as a promising pathway toward quantum advantage for real-world industrial applications.

Many scientific, technological, and industrial applications involve finding the best configuration from a vast set of discrete possibilities, such as optimizing logistics routes, scheduling tasks, designing efficient networks, or selecting the best portfolio of financial assets. Such challenging problems can be mapped mathematically to combinatorial optimization problems where the goal is to identify the best solution from an exponentially large search space while satisfying specific constraints¹.

Therefore, solving combinatorial optimization problems is highly relevant across multiple domains, as it enables improvements in efficiency, cost reduction, and performance enhancement in fields such as artificial intelligence, telecommunications, finance, healthcare, and energy systems^{1–3}. However, the computational complexity of these problems often makes them computationally challenging on classical computers^{4,5}. With the recent developments in quantum computing, speeding up the computation of industry-relevant problems is within the reach of noisy quantum processors^{6,7}. Typically, the Hamiltonian of an analog quantum system is used to encode the problem's cost function². A promising approach to solve the problem is to use an analog quantum computing device, where an initial quantum state is evolved into the ground state of the problem Hamiltonian via an adiabatic evolution^{8,9}.

Recently, arrays of neutral atoms trapped in optical tweezers have emerged as a promising processor platform for analog quantum computing^{10–12}, besides the already available quantum annealing processor^{7,13}. These analog quantum computers make use of hundreds of atoms, where each atom serves as a qubit and harness the strongly

interacting atomic Rydberg state to generate multipartite entanglement¹⁴. Moreover, the atomic arrays can be configured so that the quantum many-body ground state natively encodes the solution of optimization problems such as the maximum independent set (MIS) via adiabatic evolution^{6,15,16}. This approach can be used to tackle industrially relevant optimization problems³. A recent proposal demonstrates how to solve non-native combinatorial optimization problems on this processor¹⁷. However, in this finite-time adiabatic evolution, non-adiabatic errors are not avoidable due to the limited coherence time of the processor. The errors result in reduced computation fidelity. One way to address this challenge is by finding optimal scheduling functions to describe the adiabatic evolution¹⁸, although this can be resource-demanding and would require multiple iterations on the processor. An alternative way to circumvent the non-adiabatic excitations is by using counterdiabatic (CD) protocol as introduced in^{19–22}. The main idea behind CD protocols is to introduce an additional term to the fast-evolving adiabatic Hamiltonian to suppress the transition between eigenstates. However, the application of these initial proposals for CD protocols suffered from the difficulty in calculating the exact CD terms for large systems. Moreover, the required knowledge of instantaneous eigenstates to obtain the CD terms hindered its applications in adiabatic quantum computing (AQC). There have been several attempts to overcome this challenge²³. Notably, a proposal for a variational CD protocol^{24,25} represents significant progress in this direction. This approach offers a method to construct approximate CD terms variationally, without requiring knowledge of the Hamiltonian spectra. In this regard, several theoretical advancements have

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been made to improve this protocol^{26,27}, alongside experimental realizations on both digital and analog quantum processors^{28–33}. Additionally, digital-analog methods have recently been proposed³⁴, where sequential applications of digital and analog operations lead to targeting a wide range of Hamiltonians and use cases in a short depth circuit.

In this work, we introduce a tailored method to enhance the performance of current analog quantum processors by applying analog counterdiabatic quantum computing (ACQC) techniques, specifically designed for direct implementation on the neutral atom quantum computing platform. Our method focuses on minimizing non-adiabatic errors through the introduction of CD terms, realized through the use of analytically calculated scheduling functions that control the amplitude, detuning, and phase of the driving laser used in neutral atom quantum computing experiments. This approach significantly improves the fidelity of the computation in comparison to standard adiabatic protocols. Recognizing the limitations of current processor i.e., short coherence time, noise, and the lack of flexibility in the control variables, we tailor the CD protocols to accommodate these constraints. To demonstrate the effectiveness of our proposed CD protocols, we tackle an industrially relevant combinatorial optimization problem, the maximum independent set (MIS) problem featuring up to 100 nodes across several instances and benchmark our results against conventional finite-time adiabatic quantum optimization protocols executed on actual processor. The MIS problem maps to use cases such as network resilience, where identifying the largest set of non-adjacent nodes helps in designing robust communication networks. It also has applications in resource allocation, scheduling, and computational biology, where finding optimal, non-conflicting subsets is critical. This makes the applicability of the method relevant for industry. Additionally, we discuss the implementation of more advanced CD protocols on next-generation programmable neutral atom quantum processor, equipped with individual addressing capabilities.

In contrast to gate-based (digital) quantum processors, where CD protocols are typically implemented through sequences of discrete gates, our approach leverages the fully analog dynamics of neutral atom platforms. In a digital processor, CD terms can be engineered through gate sequences that approximate a wide variety of Hamiltonians, which provides flexibility and universality but typically at the cost of deeper circuits and reduced solution quality on current noisy devices^{28,30}. By contrast, in an analog setup, such as neutral atom platforms, the control is continuous. This allows one to target specific classes of problems with higher-quality solutions within short coherence times, though with less generality than digital schemes. In this sense, the two approaches are complementary: digital implementations offer flexibility, while analog implementations, such as ACQC, provide efficient and high-fidelity performance for tailored problem classes.

Results

We demonstrate how to enhance the capability of a neutral atom processor with ground-Rydberg qubits^{35,36} by deploying the ACQC protocol to solve the MIS problem. For this purpose, we calculated the counterdiabatic potential analytically, taking into account the processor's controllability, which includes one-body Pauli terms. This tailored approach compensates for the non-adiabatic transitions of the driving part of the ground-Rydberg qubit system. We then show a way to directly implement the CD protocol on the neutral atom processor through well-designed scheduling functions, including the CD coefficients.

Hardware implementation of ACQC

The ground-Rydberg qubits are described by the following Hamiltonian

$$\frac{H_{\text{Ryd}}(t)}{\hbar} = \underbrace{\frac{\Omega(t)}{2} \left[\cos \varphi(t) \sum_{i=1}^N \sigma_i^x - \sin \varphi(t) \sum_{i=1}^N \sigma_i^y \right]}_{H_{\text{drive}}} - \underbrace{\Delta(t) \sum_{i=1}^N \sum_{j=1}^N n_i}_{H_{\text{int}}} + \sum_{i < j} J_{ij} n_i n_j, \quad (1)$$

where $\Omega(t)$ is the Rabi frequency, $\Delta(t)$ is the detuning of the two-photon transition, $\varphi(t)$ is the phase of the laser, $n_i = |1\rangle_i \langle 1| = (1 - \sigma_i^z)/2$, and $J_{ij} = C_6/r_{ij}^6$ is the interaction energy as a function of the dispersion coefficient C_6 that depends on the chosen Rydberg state, and the distance between two atoms i and j . To simplify the calculation, we assume $\hbar = 1$ in this work.

Note that (piecewise) linear ramping scheduling functions are the easy choice to control the Rydberg system while fulfilling boundary conditions: $\Omega(0) = 0$, $\Delta(0) = -\Delta_0$, $\Omega(T) = 0$, and $\Delta(T) = \Delta_0$, where Ω_0 and Δ_0 are the maximum detuning parameters satisfying the experimental limitation. However, generally, they are not efficient at solving the MIS problem for a shorter evolution time. Therefore, in the rest of the manuscript, we use the linear control functions as a baseline AQC protocol to solve MIS problems. Beyond that, a set of smooth scheduling functions for $\Omega(t)$, $\Delta(t)$ and $\varphi(t)$ are as follows,

$$\Omega(t) = \Omega_0 \sin^2\left(\frac{\pi}{2} \sin\left(\frac{\pi t}{T}\right)\right), \quad (2)$$

$$\Delta(t) = -\Delta_0 \cos\left(\frac{\pi t}{T}\right), \quad (3)$$

$$\varphi(t) = 0, \quad (4)$$

which shows better performance on average compared with the linear protocol. Based on the smooth AQC protocol in Eq. (2–4), ACQC protocol is calculated and benchmarked to further improve the results. Besides, since there is no strict constraint on $\varphi(t)$ to solve MIS, we start with a simple case $\varphi = 0$, and the Rydberg Hamiltonian in Eq. (1) becomes

$$H_{\text{Ryd}}(t) = \frac{\Omega(t)}{2} \sum_i \sigma_i^x - \Delta(t) \sum_i n_i + \sum_{i < j} J_{ij} n_i n_j, \quad (5)$$

and the corresponding CD terms obtained analytically from Eq. (15) following the Methods section is

$$H_{\text{CD}}(t)|_{T \rightarrow 0} = -\frac{\Omega \dot{\Delta} - \Delta \dot{\Omega}}{2(\Omega^2 + \Delta^2)} \sum_i \sigma_i^y. \quad (6)$$

After adding the above CD terms into Eq. (5), the new control functions of Rydberg Hamiltonian in Eq. (1) become

$$\tilde{\Omega}(t) = \sqrt{g_1^2 + g_2^2}, \quad (7)$$

$$\tilde{\Delta}(t) = \Delta, \quad (8)$$

$$\tilde{\varphi}(t) = -\phi, \quad (9)$$

where $g_1(t) = \Omega$, $g_2(t) = -(\Omega \dot{\Delta} - \Delta \dot{\Omega})/(\Omega^2 + \Delta^2)$, and $\phi(t) = \text{atan2}(g_2, g_1)$. Considering processor implementation, CD scheduling functions can be normalized $\Omega_0 \tilde{\Omega}(t) / \max(\tilde{\Omega}(t))$ to reach the same maximum amplitude of the boundary condition. Finally, these CD scheduling functions are used to tackle MIS problems on neutral atom quantum processors. We note that the smooth scheduling functions used here are not globally optimal but represent a practical and experimentally feasible improvement over the simple linear ramp. While further optimization could be achieved with variational or numerical methods, such approaches require additional iterations and resources. Our ACQC protocol is built on top of these smooth schedules to improve the performance without the heavy overhead of variational optimization, and demonstrate robustness to the choice of baseline functions.

Performance benchmarking of ACQC with numerical simulations

In order to validate the performance of ACQC, we consider MIS problems on unit-disk graphs as a use case. For smaller graphs, we first perform noiseless simulations using the nearest-neighbor blockade subspace approximation. In Fig. 1(A), we show an example of a King's graph containing 15 nodes/qubits, which is mapped to the atom positions on a 2-dimensional grid. We apply both AQC and ACQC protocols to solve this MIS problem, and to validate the solution, we consider approximation ratio which is the ratio between the mean energy of the output of the protocol and the lowest energy, i.e., the energy of the solution, with the following formula $r = E_{\text{Mean}}/E_{\text{MIS}}$ as a metric where E_{MIS} is the ground state energy encoding the MIS solution and E_{Mean} is the mean energy of the system at the final time of the evolution. For small- and medium-sized graphs used in our simulations, E_{MIS} is computed exactly with classical solvers. For larger experimental graphs (up to 100 nodes), we obtain the MIS solution using the MIS function from the NetworkX library, which in practice provides a reliable approximation of the ground-state energy for these graph sizes. For significantly larger graphs, computing E_{MIS} exactly becomes computationally challenging, but within the scope of this work, it remains tractable.

In Fig. 1B, we plot the approximation ratio for different evolution times T for both AQC and ACQC protocols. We can see that the smooth AQC method shows better results than the linear AQC, moreover, our ACQC protocol improves on top of the smooth one. For a shorter experimental evolution time, taking $T = 1 \mu\text{s}$ as an example, the approximation ratio is improved from 0.867 for linear baseline AQC and 0.878 for smooth AQC to 0.944 for ACQC. Therefore, ACQC consistently outperforms both smooth and linear, with the most significant advantage at shorter evolution times, with upto 26% better than Linear and 7% better than Smooth at early times.

For longer evolution time, the performance of AQC and ACQC starts to converge, which showcases the adiabatic time limit as expected. In addition to the improvement in approximation ratio, ACQC also enhances the success probability (SP), defined as the fraction of runs yielding a bit-string corresponding to an exact MIS solution. At short evolution times, diabatic excitations reduce SP significantly for standard AQC, whereas the counterdiabatic corrections in ACQC suppress these transitions and keep the system closer to the ground state. Considering the same short evolution time, $T = 1 \mu\text{s}$, ACQC protocol shows clear improvement $\text{SP}^{\text{ACQC}} = 60\%$ compared with $\text{SP}^{\text{linear}} = 27.8\%$ and $\text{SP}^{\text{smooth}} = 31.4\%$.

To show statistical evidence of the superiority of ACQC over AQC, we consider 100 randomly generated graphs corresponding to MIS problems for the number of nodes between 10 and 16. The graphs are chosen to be King's graphs as the nodes of those graphs are placed on a grid, which makes them easier to implement on neutral atom processor. They are generated by choosing the size of the grid, the number of nodes, and the probability of having a node at a given crossing. The positions are then randomly generated to create a different topology for each seed fed into the generator. In Fig. 2, we fix the evolution time $T = 1 \mu\text{s}$ and plot approximation ratio vs the increasing number of nodes/qubits for the comparison of both AQC and ACQC protocols. We observe an enhancement on average by 10% with ACQC. While the difference in approximation ratio appears to narrow with increasing number of nodes in Fig. 2, this effect is largely due to statistical fluctuations in the adiabatic protocols, which depend sensitively on the random graph instances. By contrast, ACQC results remain consistently robust across instances. For example, certain graphs show better performance for adiabatic schedules, yet ACQC maintains stable improvement. This robustness is further validated in the 100-node experimental demonstration (Fig. 3C) in the next section, where ACQC preserves a 6–8% advantage compared with the smooth AQC protocol at short evolution time.

Performance benchmarking of ACQC on neutral atom processor

Following the clear improvement of ACQC protocol demonstrated by the simulation results, we tackle larger MIS problems on actual processor. We utilized the cloud-based neutral atom processor platform provided by QuEra and Pasqal's neutral atom processor platform as a testbed to evaluate the performance of the ACQC protocol. We employed the Aquila device consisting of 256 qubits, to solve an instance of the MIS problem on a 100 nodes/qubits King's graph, as shown in Fig. 3A.

The experimental results are compared across ACQC, an AQC protocol with smooth functions, and a commonly used AQC protocol with linear scheduling functions. Similar to the simulations, the scheduling functions for the smooth AQC are chosen as outlined in Eqs. (2–4), the ACQC protocol is calculated based on the same scheduling functions. Since linear scheduling function is a common choice on neutral atom processor, it serves as a reference here. Having previously computed the optimal solution of the MIS problem using a classical algorithm^{37,38}, we can compare the optimality of the solution found through each protocol. We scan three different evolution times for each protocol, and each computation was performed with 1000 shots. In Fig. 3D, we compare the minimum energy

Fig. 1 | Comparison of simulation results for solving a 15-node/qubit MIS problem. A A randomly generated graph with 15 nodes with one of the MIS solutions showing as nodes in brown. B The approximation ratio as a function of evolution time T for solving the MIS of the graph plotted in (A) is compared between the linear protocol (linear scheduling functions, grey dash-dotted line with circles), smooth AQC (smooth scheduling functions, orange dashed line with squares) and ACQC (calculated based on the smooth AQC protocol, solid line with upper triangles). For one example of $T = 1 \mu\text{s}$, the energy distribution analysed by using Kernel density estimation is plotted in (C). Parameters: $\Omega_0 = 15 \text{ MHz}$, $\Delta_0 = 17 \text{ MHz}$, 500 shots for each evolution, and the linear protocol's Rabi frequency ramping time is $0.15 T$ and detuning ramping time is $0.7 T$. The atoms are also placed on a square grid of length $5.5 \mu\text{m}$, ensuring that atoms from the same square unit and on the diagonal are within the Rydberg blockade range from each other. We choose C_6 for Rb-87 and a Rydberg state of principal quantum number 70, $C_6 = 862690 \times 2\pi \text{ MHz } \mu\text{m}^6$.

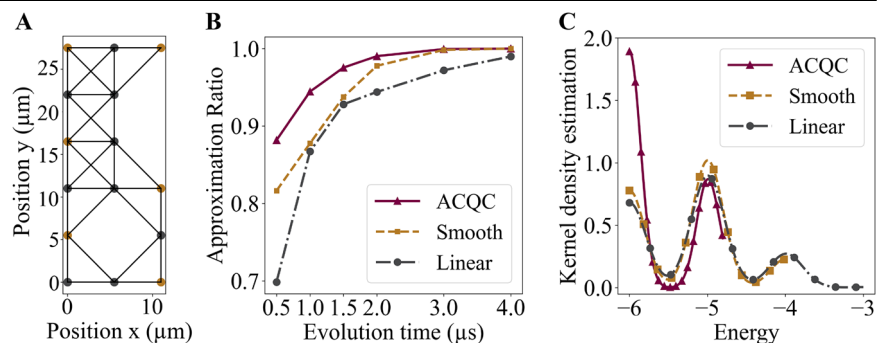


Fig. 2 | Statistical evidence of ACQC enhancement on simulator. Noiseless simulation results for the approximation ratio for evolution time $T = 1 \mu\text{s}$ for different numbers of nodes per qubits. For each number of qubit box plot, we simulate 100 randomly generated graphs for the statistical evidence. The comparison is between the ACQC, smooth AQC and linear AQC protocols with the same parameters as in Fig. 1.

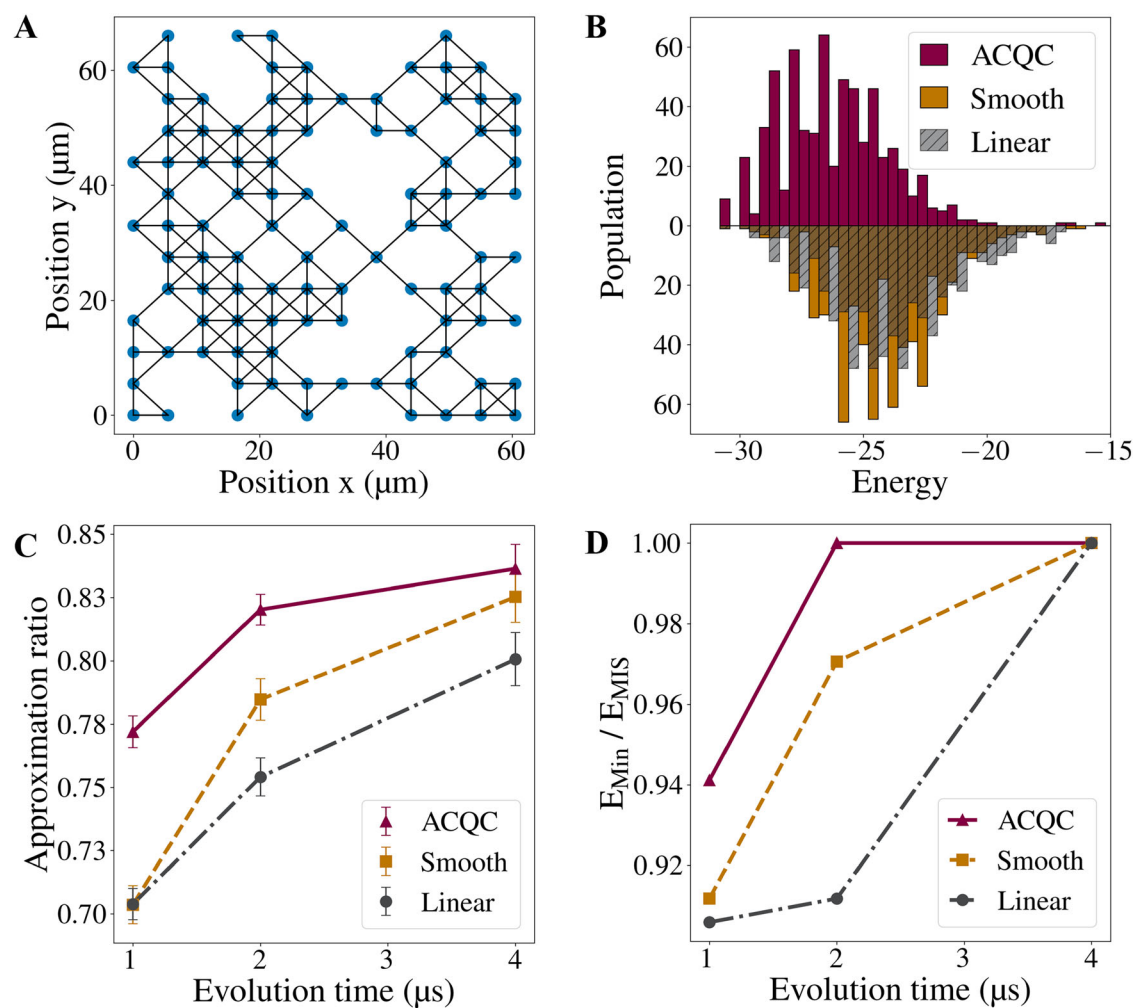
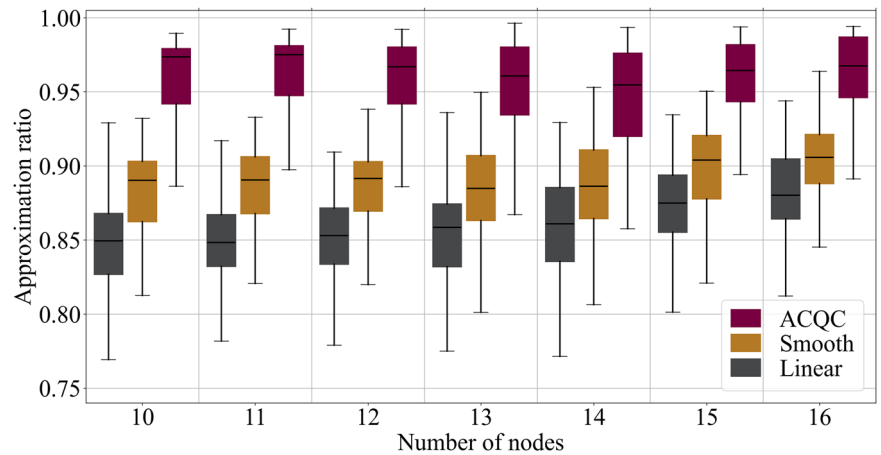


Fig. 3 | Implementation on QuEra's quantum processor. Experimental results obtained from solving MIS for a 100 nodes graph using QuEra's cloud-accessed Aquila quantum processor using linear scheduling functions (linear), smooth chosen scheduling functions (smooth), and our ACQC protocol calculated from the smooth AQC protocol scheduling functions. **A** Implementation of the graph onto the atomic register. The connections are drawn when two atoms are

separated by less than the blockade radius from one another. **B** Distribution of bitstring energy for an evolution time of $1 \mu\text{s}$. **C** Evolution of the approximation ratio with different evolution time, with confidence interval. **D** Evolution of the ratio between minimum energy obtained and ground state energy or the energy of the state encoding the MIS solution with different evolution time. Parameters are the same as in Fig. 1.

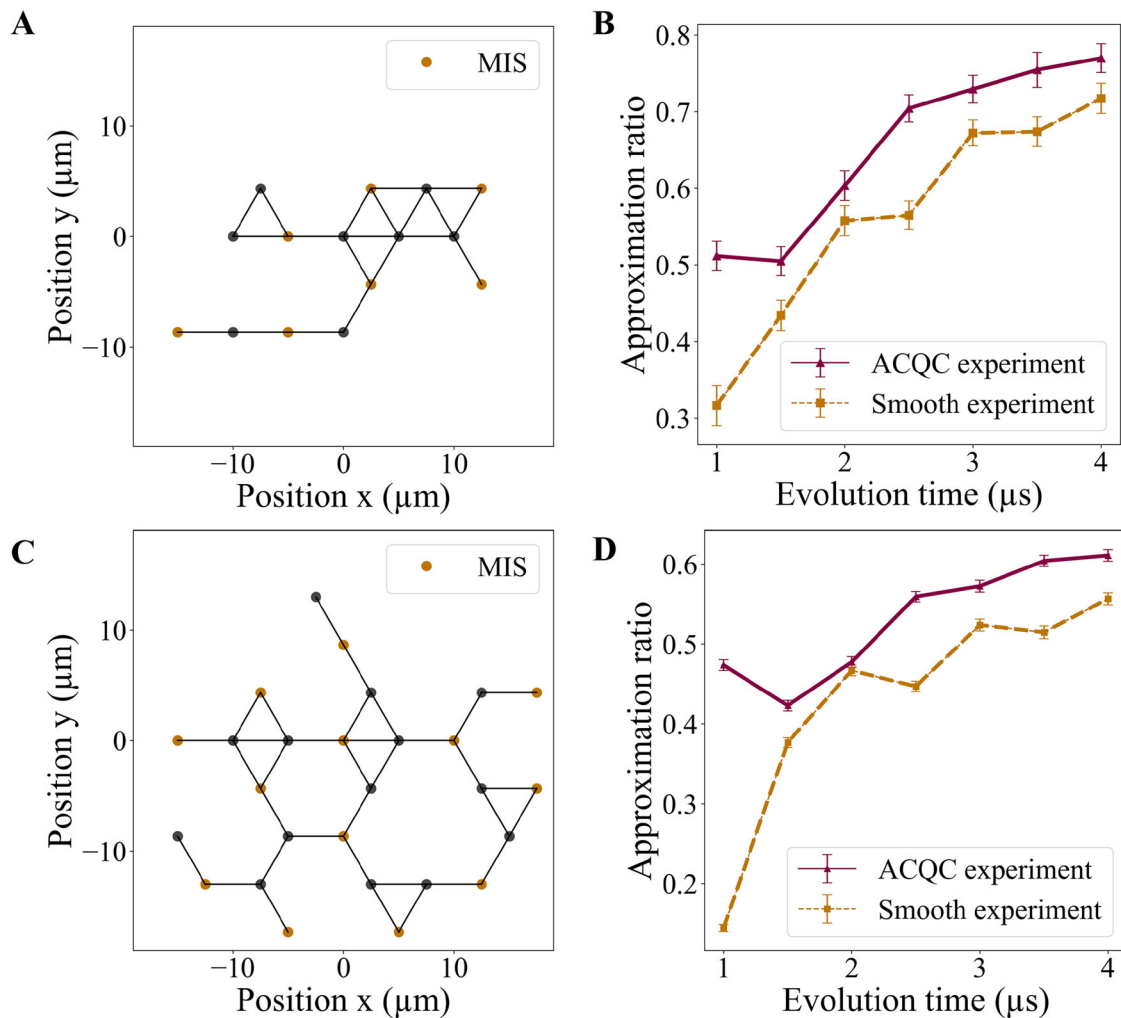


Fig. 4 | Implementation on Pasqal's quantum processors. Experimental results obtained from solving MIS for a 15 and 27 nodes/qubits graphs, using Pasqal's Orion Alpha quantum processor. We use smooth schedule functions for both ACQC and the AQC protocol. **A** Mapping of the 15 nodes/qubits graph onto the atomic register. The connections are drawn when two atoms are separated by less than the blockade radius and one solution is highlighted. **B** The approximation ratio for different evolution time for the 15 nodes graph shown in (A), with confidence

interval. **C** Mapping of the 27 nodes/qubits graph onto the atomic register. The connections are drawn when two atoms are separated by less than the blockade radius, and one MIS solution is highlighted. **D** The approximation ratio with different evolution time for the 27 nodes/qubits graph in (C), with confidence interval. Parameters are the same as in Fig. 1. Parameters: $\Omega_0 = 25.1327$ MHz, $\Delta_0 = 31.4159$ MHz, grid length = $5 \mu\text{m}$, $C_6 = 8.65722 \times 10^5$ MHz μm^6 for the Orion Alpha Rydberg state.

obtained at different evolution times with the ground-state energy whose bitstring encodes the MIS solution. We also compare the mean energy in Fig. 3C, where the error bar represents the statistical confidence interval calculated from the number of shots and the standard deviation of the distribution of bitstring's energy. As expected, for both the minimum energy and the mean energy, ACQC improves the results over AQC for a shorter time $T = 1 \mu\text{s}$ and $T = 2 \mu\text{s}$.

For longer evolution time, $T = 4 \mu\text{s}$, the approximation ratios for both the smooth AQC and ACQC are nearly identical and approach unity. However, the mean energy observed with ACQC is notably lower than that of the smooth AQC, and, as expected, significantly lower than the linear model. The distribution of energy obtained for a total evolution time of $1 \mu\text{s}$ is plotted in Fig. 3(B).

We perform similar experiments on Pasqal's Orion Alpha quantum processors, solving MIS for a 15 nodes/qubits graph and a 27 nodes/qubits graph in Fig. 4. To showcase a different implementation of our ACQC protocol, we mapped the CD protocol using strictly the Rabi frequency and the detuning without the need for the control of the phase. This is done by applying a Z -rotation on our ACQC evolution by using a unitary operator $U = \exp(i\phi\sigma^z/2)$ to control the Rydberg Hamiltonian without phase

$H^{\text{Z-rot}} = (\tilde{\Omega}^{\text{Z-rot}}(t)/2) \sum_i \sigma_i^x - \Delta^{\text{Z-rot}}(t) \sum_i n_i + \sum_{i < j} J_{ij} n_i n_j$. Therefore, the ACQC scheduling functions without the control of phase are $\tilde{\Omega}^{\text{Z-rot}}(t) = \tilde{\Omega}(t)$, $\tilde{\Delta}^{\text{Z-rot}}(t) = \tilde{\Delta}(t) + \tilde{\phi}(t)$. We compare performance for solving MIS using the same protocol as the previous paragraph by plotting the approximation ratio for different evolution times, using only the smooth AQC scheduling functions and our ACQC protocol calculated from the smooth AQC functions. Both experiments confirm the expected enhancement of ACQC. In the fast quench regime ($1 \mu\text{s}$), Panel B demonstrates an approximate 60% improvement in performance. For the 27-node graph, Panel D reveals an even greater advantage, with ACQC achieving a 5× improvement at short evolution times. Compared with variationally optimized adiabatic schedules, which require extensive numerical searches for each problem instance, ACQC achieves systematic improvements with only modest experimental overhead, since the analytic counterdiabatic corrections are derived once and directly implemented through modified control waveforms. This highlights the effectiveness of ACQC in accelerating optimization performance, particularly in rapid quench scenarios.

We note that the experimental graphs differ from the simulated King's graphs of Fig. 2 due to hardware constraints. In QuEra's Aquila device, the Rydberg state and connectivity yield dynamics comparable to our simulations, and the improvement is consistent (6–8% at 1 μ s). By contrast, Pasqal's Orion Alpha hardware encodes non-King's graphs and employs a different Rydberg state, altering the adiabatic timescale. In this case, 1 μ s lies deeper into the fast-quench regime, making a direct comparison less meaningful. At longer times, the observed improvement (5–10%) is consistent with both QuEra experiments and simulation trends, highlighting that the observed differences stem from hardware-specific constraints.

Discussion

We experimentally demonstrated the first implementation of analog counterdiabatic quantum computing (ACQC) on a neutral atom quantum processor for solving a combinatorial optimization problem. Our ACQC protocol is a general analytical method and can be deployed on current and next generation neutral atom processors. Technically, by adding the adiabatic gauge potential of the driving part of the Rydberg Hamiltonian to the system with a given set of scheduling functions, we achieve rapid and high-quality solutions for computational problems without the need for any optimization on the processor. It achieves 3 $\times$ speedup and enhanced success probabilities in fast quench regimes (1 μ s), while also optimizing the mean energy of the bitstring distribution, making it valuable for quantum sampling applications³⁹. In this ACQC implementation, the counterdiabatic protocol is derived from the adiabatic gauge potential of the Rydberg Hamiltonian in the zero-interaction limit, enabling faster solutions. ACQC outperforms standard AQC at short evolution times, with conventional AQC matching its performance at $\sim 4 \mu$ s.

The MIS problem addressed in this work maps to a variety of practical applications, including network resilience, resource allocation, scheduling, and computational biology domains, where identifying optimal, non-conflicting subsets is crucial. This highlights the method's strong relevance for industry. Furthermore, the proposed protocol can be extended to next-generation processors that support digital operations, enabling digital-analog quantum computing to tackle a broader class of Hamiltonians. Recent experimental advances have demonstrated the feasibility of this approach in superconducting processors⁴⁰, along with proposals for implementing counterdiabatic dynamics in trapped-ion systems³⁴.

Moreover, the development of tailored CD techniques opens new avenues for fast analog quantum simulations, for example, in the study of critical dynamics of two-dimensional spin systems with high precision⁴¹. Although recent demonstrations on digital processors with hundreds of qubits highlight this capability⁴¹, analog platforms combined with optimized CD protocols hold promise to deliver superior solution quality.

Our work shows that the ACQC necessitates only the dynamic manipulation of Rabi frequency and detuning, alongside the dynamical adjustment of the driving laser's phase. We also propose an alternative implementation that does not require the controllability of the laser phase. This compatibility enables us to conduct trials via cloud access to QuEra's Aquila device as well as using Pasqal's Orion Alpha device, leveraging their advanced capabilities for comprehensive evaluation.

Finally, the fast-quench solution feature can be used in sequential processes where one part of the computation time can be used to prepare a given state and the other one to perform a high-fidelity computation within the coherence time of the processor.

While our present implementation demonstrates ACQC on up to 100 qubits without requiring local addressability, future scalability toward arbitrary problem instances will benefit from hardware advances such as local detuning, single-qubit addressability, and extended coherence times. The ACQC protocol is designed primarily to provide rapid, high-quality solutions within the limited coherence window of current devices, rather than to mitigate noise directly. Nevertheless, reductions in device noise will naturally improve the overall quality of the solutions. These features would allow ACQC protocols to incorporate enhanced counterdiabatic protocols, enabling broader applicability while maintaining robustness. Importantly,

since ACQC already provides 5–10% improvement in approximation ratios at short evolution times, the experimental requirements for demonstrating quantum advantage (e.g., coherence time, control fidelity) are relaxed compared to standard adiabatic protocols, making ACQC a realistic candidate for near-term advantage on next-generation neutral atom processors.

Methods

Solving combinatorial optimization problem and adiabatic quantum computing method

Many combinatorial optimization problems, including the maximum independent set (MIS), traveling salesman problem, and quadratic assignment problem, can be efficiently encoded as a quadratic unconstrained binary optimization (QUBO) problem². The usual neutral atom dynamics, specifically suited for tackling such optimization problems, enable the mapping of problems onto an Ising Hamiltonian. This results in a graph problem with long-range interactions among neighboring atoms, manifesting the Rydberg blockade phenomenon, preventing simultaneous excitation of adjacent atoms to the Rydberg state.

Focusing on the current neutral atom platform with ground-Rydberg qubits, this work centers on solving the MIS problem on a unit disk graph. Mathematically, the MIS problem is defined as finding a set S of vertices in a graph $G = (V, E)$ such that no two vertices in S share an edge, and S is the largest set satisfying this condition. In unit disk graphs, each vertex v represents a disk of uniform radius, with an edge (u, v) existing between two vertices if and only if the corresponding disks overlap. This spatial property of unit disk graphs correlates well with the operational dynamics of neutral atom quantum processors, which can exploit their Rydberg blocked phenomena to efficiently realize this problem. The cost function corresponding to this problem is given by

$$H(x) = A \sum_{(u,v) \in E} x_u x_v - B \sum_{v \in V} x_v, \quad (10)$$

where x_v are binary variables, which take the value 1 if vertex v is included in the independent set S , and 0 otherwise. The first sum penalizes any edges where both vertices are included in the set S (i.e., it enforces the independence condition). The second sum rewards the inclusion of vertices in the set S . A and B are constants where $A > B$ to ensure that the penalty for violating the independent set condition is higher than the reward for including additional vertices. The objective is to find a configuration of x_v that minimizes $H(x)$. This minimization problem with quadratic terms can be mapped to finding the ground state of an Ising Hamiltonian. This can be tackled using AQC methods.

Adiabatic quantum computing is a well-known approach for solving combinatorial optimization problems, especially when using analog quantum computing processor. In this method, one begins by selecting an initial Hamiltonian H_i , whose ground state is both known and easy to prepare. The system is then adiabatically evolved towards the problem Hamiltonian H_p by slowly changing the driving terms as defined by a time-dependent Hamiltonian $H(t) = f(t)H_i + g(t)H_p$. For sufficiently slow evolution, the adiabatic theorem ensures that the system remains close to its ground state throughout the process. In this case, the wave function of the system follows the instantaneous eigenstates of the Hamiltonian while the optimization solutions are encoded to be the ground state of the final Hamiltonian, which is the target state. However, the adiabatic evolution requires long computation times, which are limited by the experiment, for example, the coherence time of the neutral atom system. A nonadiabatic evolution or the noise of the system can lead to excitations in the energy spectrum and can reduce the target-state fidelity, in other words, the success probability. Therefore, we propose analog counterdiabatic quantum computing method.

The ground-Rydberg Hamiltonian

Consider the Hamiltonian of neutral atom platform using ground-Rydberg qubits in Eq. (1). Since the initial state of this processor is $|0\rangle^{\otimes N}$, the initial conditions read $\Omega(0) = 0$ and $\Delta(0)$ is negative. At the final evolution time $t = T$, the Rabi frequency is back to zero and the ground state of the final

Hamiltonian $H_{\text{Ryd}}(T)$ encodes the solution of the combinatorial optimization problem, for example minimizing the cost function in Eq. (10). Combining the initial and target constraints, one obtains the boundary conditions of the control functions as follows

$$\Omega(0) = 0, \Delta(0) = -\Delta_1, \quad (11)$$

$$\Omega(T) = 0, \Delta(T) = \Delta_2, \quad (12)$$

where $-\Delta_1$ and Δ_2 are the minimal (negative) and maximal (positive) experimental limitations of the detuning. Analytically, the minimal eigenvalue of the initial Hamiltonian is 0, and the second minimal eigenvalue is $\Delta_1 > 0$. Therefore, choosing a larger value of Δ_1 can enlarge the gap between the initial ground state and the initial first excited state. For the final constraint, Δ_2 can influence the minimal eigenvalue and the energy gap between the ground state and the first excited state at the final time. Depending on the specific graph and its structure, Δ_2 should be set based on the number of vertices and edges to satisfy the condition that the ground state encodes the solution of the MIS problems.

Note that no boundary condition applies to φ since its effect vanishes at initial and final evolution times due to Ω . In all standard protocols, the choice is to choose a constant phase: $\varphi = 0$.

ACQC protocol

Hereafter, we show a way to design the scheduling functions $\Omega(t)$, $\Delta(t)$, and $\varphi(t)$ to not only fulfill the boundary conditions, but also improve the success probability for a shorter computational time.

The idea of counterdiabaticity^{20,21} is to add an auxiliary Hamiltonian H_{CD} to the adiabatic Hamiltonian. This helps to guide the system more reliably to the desired state by preventing non-adiabatic transitions. Therefore, the Hamiltonian becomes

$$H_{\text{tot}}(t) = H_{\text{ad}}(t) + H_{\text{CD}}(t). \quad (13)$$

There are different ways to add CD terms. Considering a current ground-Rydberg quantum computing platform, the Ising mode interactions exist when two neighboring atoms i and j are in the Rydberg states, n_i and n_j terms in Eq. (1). In the case of many-body systems, the exact adiabatic gauge potential of the dynamic system cannot be found or the energy spectrum is too expensive to calculate, obviously a nested commutator CD terms protocol²⁴ could be a possible solution which is a variational method to search for an approximation of adiabatic gauge potential. However, it requires additional many-body terms, which are currently not available to be added to analog neutral atom quantum computing processor.

To find a way around, we develop an ACQC method which does not require additional many-body interaction terms added to the quantum computing system. This can be directly implemented on the current neutral atom quantum processors without optimization or post-processing on processor.

To ensure that the system follows the desired adiabatic path and reaches the ground state of a Hamiltonian $H_{\text{ad}}(t)$ at the final time $t = T$, the constraint for $H_{\text{CD}}(t)$ to be the solution of the adiabatic gauge potential²⁴ of $H_{\text{ad}}(t)$ is

$$[\partial_t H_{\text{ad}} - i[H_{\text{ad}}, H_{\text{CD}}], H_{\text{ad}}] = 0. \quad (14)$$

To avoid introducing extra many-body terms beyond $\sigma_i^z \sigma_j^z$ terms of the Rydberg system, an efficient solution is to search for the counterdiabaticity of the independent spins under the control field where the Hamiltonian is the driving part of Rydberg Hamiltonian in Eq. (1), $H_{\text{ad}}(t) = H_{\text{drive}}(t)$. Then, the adiabatic gauge potential of $H_{\text{ad}}(t)$ in the limit of zero interactions can be easily solved by choosing the following CD ansatz

$$H_{\text{CD}}(t)|_{J \rightarrow 0} = f_x(t) \sum_i \sigma_i^x + f_y(t) \sum_i \sigma_i^y + f_z(t) \sum_i n_i, \quad (15)$$

where the general solution of the CD coefficients $f_{x,y,z}$ in Eq. (15) can be analytically calculated directly through Eq. (14) as

$$f_x(t) = -\frac{\Omega \dot{\Delta} - \dot{\Delta} \Omega}{2(\Omega^2 + \Delta^2)} \sin \varphi + \frac{\Omega \Delta \dot{\varphi}}{2(\Omega^2 + \Delta^2)} \cos \varphi, \quad (16)$$

$$f_y(t) = -\frac{\Omega \dot{\Delta} - \dot{\Delta} \Omega}{2(\Omega^2 + \Delta^2)} \cos \varphi - \frac{\Omega \Delta \dot{\varphi}}{2(\Omega^2 + \Delta^2)} \sin \varphi, \quad (17)$$

$$f_z(t) = \frac{\Omega^2 \dot{\varphi}}{\Omega^2 + \Delta^2}. \quad (18)$$

Finally, the total Hamiltonian with CD terms in Eq. (13) should be implemented through the Rydberg Hamiltonian $\tilde{H}_{\text{Ryd}}(t) = H_{\text{tot}}(t)$ with the updated scheduling functions as follows:

$$\tilde{H}_{\text{Ryd}}(t) = \frac{\tilde{\Omega}(t)}{2} \left[\cos \tilde{\varphi}(t) \sum_i \sigma_i^x - \sin \tilde{\varphi}(t) \sum_i \sigma_i^y \right] - \tilde{\Delta}(t) \sum_i n_i + \sum_{i < j} J_{ij} n_i n_j. \quad (19)$$

Therefore, the counterdiabatic scheduling functions are calculated as

$$\tilde{\Omega}(t) = \sqrt{g_1^2 + g_2^2}, \quad (20)$$

$$\tilde{\Delta}(t) = \Delta - \frac{\Omega^2 \dot{\varphi}}{\Omega^2 + \Delta^2}, \quad (21)$$

$$\tilde{\varphi}(t) = \varphi - \phi, \quad (22)$$

with

$$g_1(t) = \Omega \left(1 + \frac{\Delta \dot{\varphi}}{\Omega^2 + \Delta^2} \right), \quad (23)$$

$$g_2(t) = -\frac{\Omega \dot{\Delta} - \dot{\Delta} \Omega}{\Omega^2 + \Delta^2}, \quad (24)$$

$$\phi(t) = \text{atan2}(g_2, g_1), \quad (25)$$

where $\text{atan2}(y, x)$ returns the four-quadrant inverse tangent of y and x . Obviously, the scheduling functions $\Omega(t)$, $\Delta(t)$, and $\varphi(t)$ are free to be chosen with respect to the boundary conditions in Eqs. (11 and 12) and the experimental limitations. Once the scheduling functions are set, the ACQC control protocol in Eqs. (20–25) are designed and implemented on commercial neutral atom processor.

Data availability

The datasets generated and/or analyzed during the current study, including the simulation outputs, processed experimental results, and benchmarking data, are available from the corresponding authors upon reasonable request. Due to the use of proprietary quantum hardware and commercial platform restrictions (QuEra's Aquila and Pasqal's Orion Alpha), raw device-level data and control parameters are subject to confidentiality agreements with the hardware providers and cannot be publicly shared. Derived data supporting the findings of this study are available from the corresponding authors upon reasonable request. The custom codes used for numerical simulations, data analysis, and generation of the figures in this study were developed in Python, making use of standard scientific computing libraries such as NumPy, SciPy, Matplotlib, and NetworkX for graph generation and classical solution benchmarking. The quantum experiments were executed on QuEra's Aquila and Pasqal's Orion Alpha neutral atom quantum

processors through their respective cloud-based software development kits (Bloqade SDK/Amazon Braket for QuEra, and Pulser SDK for Pasqal). Due to confidentiality agreements with the hardware providers and the proprietary nature of some code components, the complete simulation and control scripts are available from the corresponding authors upon reasonable request.

Code availability

The custom codes used for numerical simulations, data analysis, and generation of the figures in this study were developed in Python, making use of standard scientific computing libraries such as NumPy, SciPy, Matplotlib, and NetworkX for graph generation and classical solution benchmarking. The quantum experiments were executed on QuEra's Aquila and Pasqal's Orion Alpha neutral atom quantum processors through their respective cloud-based software development kits (Bloqade SDK/Amazon Braket for QuEra, and Pulser SDK for Pasqal). Due to confidentiality agreements with the hardware providers and the proprietary nature of some code components, the complete simulation and control scripts are available from the corresponding authors upon reasonable request.

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References

1. Fu, Y. & Anderson, P. W. Application of statistical mechanics to NP-complete problems in combinatorial optimization. *J. Phys. A Math. Gen.* **19**, 1605 (1986).
2. Lucas, A. Ising formulations of many NP problems. *Front. Phys.* **2**, 5 (2014).
3. Wurtz, J., Lopes, P. L. S., Gemelke, N., Keesling, A. & Wang, S.-T. Industry applications of neutral-atom quantum computing solving independent set problems. *arXiv preprint* <https://doi.org/10.48550/arXiv.2205.08500> (2022).
4. Lenstra, J. K. & Rinnooy Kan, A. H. G. Computational complexity of discrete optimization problems. *Ann. Discret. Math.* **4**, 121–140 (1979).
5. Colnari, A. et al. Heuristics from nature for hard combinatorial optimization problems. *Int. Trans. Oper. Res.* **3**, 1–21 (1996).
6. Ebadi, S. et al. Quantum optimization of maximum independent set using Rydberg atom arrays. *Science* **376**, 1209–1215 (2022).
7. King, A. D. et al. Quantum critical dynamics in a 5000-qubit programmable spin glass. *Nature* **617**, 61–66 (2023).
8. Farhi, E., Goldstone, J., Gutmann, S. & Sipser, M. Quantum computation by adiabatic evolution. *arXiv preprint* <https://doi.org/10.48550/arXiv.quant-ph/0001106> (2000).
9. Albash, T. & Lidar, D. A. Adiabatic quantum computation. *Rev. Mod. Phys.* **90**, 015002 (2018).
10. Browaeys, A. & Lahaye, T. Many-body physics with individually controlled Rydberg atoms. *Nat. Phys.* **16**, 132–142 (2020).
11. Scholl, P. et al. Quantum simulation of 2D antiferromagnets with hundreds of Rydberg atoms. *Nature* **595**, 233–238 (2021).
12. Ebadi, S. et al. Quantum phases of matter on a 256-atom programmable quantum simulator. *Nature* **595**, 227–232 (2021).
13. Brooke, J., Bitko, D., Rosenbaum, T. F. & Aeppli, G. Quantum annealing of a disordered magnet. *Science* **284**, 779–781 (1999).
14. Morgado, M. & Whitlock, S. Quantum simulation and computing with Rydberg-interacting qubits. *AVS Quantum Sci.* **3**, 023501 (2021).
15. da Silva Coelho, W., D'Arcangelo, M. & Henry, L.-P. Efficient protocol for solving combinatorial graph problems on neutral-atom quantum processors. *arXiv preprint* <https://doi.org/10.48550/arXiv.2207.13030> (2022).
16. Wurtz, J., Lopes, P., Gemelke, N., Keesling, A. & Wang, S.-T. Industry applications of neutral-atom quantum computing solving independent set problems. *arXiv preprint* <https://doi.org/10.48550/arXiv.2205.08500> (2022).
17. Wurtz, J., Sack, S. & Wang, S.-T. Solving non-native combinatorial optimization problems using hybrid quantum–classical algorithms. *IEEE Trans. Quantum Eng.* **5**, 1–14 (2024).
18. Finžgar, J. R., Schuetz, M. J. A., Brubaker, J. K., Nishimori, H. & Katzgraber, H. G. Designing quantum annealing schedules using Bayesian optimization. *Phys. Rev. Res.* **6**, 023063 (2024).
19. Demirplak, M. & Rice, S. A. Adiabatic quantum computation-(title omitted in source). *J. Phys. Chem. A* **107**, 9937–9945 (2003).
20. Berry, M. V. Transitionless quantum driving. *J. Phys. A Math. Theor.* **42**, 365303 (2009).
21. Chen, X., Lizuain, I., Ruschhaupt, A., Guéry-Odelin, D. & Muga, J. G. Shortcut to adiabatic passage in two- and three-level atoms. *Phys. Rev. Lett.* **105**, 123003 (2010).
22. Del Campo, A. Shortcuts to adiabaticity by counterdiabatic driving. *Phys. Rev. Lett.* **111**, 100502 (2013).
23. Saberi, H., Opatrný, T., Mølmer, K. & del Campo, A. Adiabatic tracking of quantum many-body dynamics. *Phys. Rev. A* **90**, 060301 (2014).
24. Sels, D. & Polkovnikov, A. Minimizing irreversible losses in quantum systems by local counterdiabatic driving. *Proc. Natl. Acad. Sci. USA* **114**, E3909–E3916 (2017).
25. Claeys, P. W., Pandey, M., Sels, D. & Polkovnikov, A. Floquet-engineering counterdiabatic protocols in quantum many-body systems. *Phys. Rev. Lett.* **123**, 090602 (2019).
26. Takahashi, K. & del Campo, A. Shortcuts to adiabaticity in Krylov space. *Phys. Rev. X* **14**, 011032 (2024).
27. Nakahara, M. Counterdiabatic formalism of shortcuts to adiabaticity. *Philos. Trans. R. Soc. A* **380**, 20210272 (2022).
28. Hegade, N. N. et al. Shortcuts to adiabaticity in digitized adiabatic quantum computing. *Phys. Rev. Appl.* **15**, 024038 (2021).
29. Hegade, N. N., Chen, X. & Solano, E. Digitized counterdiabatic quantum optimization. *Phys. Rev. Res.* **4**, L042030 (2022).
30. Chandarana, P., Hegade, N. N., Montalban, I., Solano, E. & Chen, X. Digitized counterdiabatic quantum algorithm for protein folding. *Phys. Rev. Appl.* **20**, 014024 (2023).
31. Guan, H. et al. Single-layer digitized-counterdiabatic quantum optimization for p -spin models. *Quantum. Sci. Technol.* **10**, 015006 (2025).
32. Gomez Cadavid, A., Montalban, I., Dalal, A., Solano, E. & Hegade, N. N. Efficient DCQO algorithm within the impulse regime for portfolio optimization. *Phys. Rev. Appl.* **22**, 054037 (2024).
33. Hayasaka, H., Imoto, T., Matsuzaki, Y. & Kawabata, S. A general method to construct mean field counterdiabatic driving for a ground state search. *arXiv preprint* <https://doi.org/10.48550/arXiv.2305.08352> (2023).
34. Kumar, S. et al. Digital–analog counterdiabatic quantum optimization with trapped ions. *Quantum Sci. Technol.* **10**, 015023 (2025).
35. Lukin, M. D. et al. Dipole blockade and quantum information processing in mesoscopic atomic ensembles. *Phys. Rev. Lett.* **87**, 037901 (2001).
36. Saffman, M. & Walker, T. G. Analysis of a quantum logic device based on dipole–dipole interactions of optically trapped Rydberg atoms. *Phys. Rev. A* **72**, 022347 (2005).
37. NetworkX Developers. Maximum independent set function in networkx (2025). Used to obtain an approximate MIS solution for a given graph G .
38. Boppana, R. & Halldórsson, M. M. Approximating maximum independent sets by excluding subgraphs. *BIT Numer. Math.* **32**, 180–196 (1992).
39. da Silva Coelho, W., Henriët, L. & Henry, L.-P. Quantum pricing-based column-generation framework for hard combinatorial problems. *Phys. Rev. A* **107**, 032426 (2023).
40. Andersen, T. I. et al. Thermalization and criticality on an analogue–digital quantum simulator. *Nature* **638**, 79–85 (2025).

41. Visuri, A.-M., Gomez Cadavid, A., Bhargava, B. A., Romero, S. V. & Grabarits, A. Digitized counterdiabatic quantum critical dynamics. *arXiv preprint* <https://doi.org/10.48550/arXiv.2502.15100> (2025).

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Author contributions

Q.Z conducted primary research, data analysis, and paper writing. N.N.H contributed in the development of the idea and writing the paper. A.G.C provided support in analytical calculations that lead to validate the method in the hardware. L.L supported in hardware implementation. S.K supported in improving the paper quality and communications. J.T and S.P provided technical knowledge about neutral atoms and guided the hardware implementation. E.S guided the project. L.H provided hardware support, and E.M provided overall lead, paper writing, results, and analysis.

Competing interests

The authors declare no competing interests.

Additional information

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